Brooke Kotten

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Education

University of Michigan *August 2024-Present*
PhD, Astronomy, expected 2029 *Ann Arbor, MI*

University of Wisconsin-Madison *September 2021-May 2024*
B.S., Astronomy-Physics (Honors) and Physics *Madison, WI*

Saint Cloud State University *August 2019-May 2021*
A.A., Liberal Arts, *summa cum laude* *Saint Cloud, MN*
Concurrent with high school diploma

Research Experience

Graduate Student at the University of Michigan *August 2024-Present*
with Dr. Mary Anne Limbach (University of Michigan)
Searching for transiting exosatellites in JWST time series observations of substellar worlds and investigating the detectability of exo-Ios.

Lamat REU & UW-Madison Senior Honors Thesis *June 2023-Present*
with Professor Melinda Soares-Furtado (UW-Madison) and Professor Enrico Ramirez-Ruiz (UCSC)
Investigated the lithium enrichment in a planetary engulfment candidate.

Bolder Solar Alliance REU & Student Researcher at CIRES *May 2022-June 2023*
with Dr. Courtney Peck and Dr. Janet Machol (CIRES/NOAA/NCEI)
Improved GOES-R's X-Ray Sensor (XRS) location algorithm with new calculated parameters and a correction for near-simultaneous solar flares. My location correction method is used operationally to improve the real-time flare locations from XRS.

Undergraduate Research Scholars Program *October 2021-May 2022*
with Dr. Ralf Kotulla (UW-Madison)
Investigated star formation on the outskirts of spiral galaxies.

Publications

First Author Publications

2. **Kotten, B.**, Limbach, M. A., Vos, J.; et al. "On the Detectability of Volcanic Exo-Ios that May Fuel Auroras on Super-Jupiters." 2026, *submitted to AAS journals.*

1. **Kotten, B.**, Soares-Furtado, M., Yarza, R.; et al. "Evaluating the Lithium Abundance of a Subgiant Star with a Brown Dwarf Companion: A Planetary Engulfment Candidate." 2026, *in review with AAS Journals*

Co-Author Publications

6. Machol, J., Codrescu, S., Peck, C.; et al. **incl. Kotten, B.**. "GOES-R Series X-Ray Sensor (XRS): 2. On-orbit measurements and calibrations." 2026, *submitted to AGU journals.*

5. Limbach, M.A., Dacus, B., **Kotten, B.**; et al. "Exomoons and Exorings with the Habitable Worlds Observatory II: Finding 'Endor' with Lunar Eclipses." 2026, The Astronomical Journal, 171, 4.

4. Householder, A., Limbach, M. A., Biller, B., **Kotten, B.**; et al. "Sensitivity to Sub-Io-sized Exosatellite Transits in the MIRI LRS Lightcurve of the Nearest Substellar Worlds." 2025, The Astrophysical Journal Letters, 992, 1.

3. Wilson, M. J., Limbach, M. A., Skemer, A. J.; et al. **incl. Kotten, B.**. "A Deep Search for Exomoons Around WISE 0855 With JWST." 2025, The Astronomical Journal, 170, 6.

2. Schulte, J., Rodriguez, J. E., Latham, D. W., Sheilds, J. V., Vowell, N., Soares-Furtado, M., **Kotten, B.**; et al. "Migration and Evolution of giant ExoPlanets (MEEP) II: Super-Jupiters and Lithium-rich Host Stars." 2025, Monthly Notices of the Royal Astronomical Society, 543, 1.

1. Vowell, N., Rodriguez, J. E., Latham, D. W., Quinn, S. N., Schulte, J.; et al. **incl. Kotten, B.**. "Eleven New Transiting Brown Dwarfs and Very-low-mass Stars from TESS." 2025, The Astronomical Journal, 170, 2.

Science Presentations

Nov 2025 **Great Lakes Exoplanet Area Meeting**, Madison, WI [Talk]
July 2025 **OWL Exoplanet Summer Program**, Santa Cruz, CA [Talk]
June 2024 **244th American Astronomical Society Summer Meeting**, Madison, WI [Poster]
Apr 2024 **UW-Madison Undergraduate Symposium**, Madison, WI [Poster]
Jan 2024 **243rd American Astronomical Society Winter Meeting**, New Orleans, LA [Poster]
Dec 2023 **UW-Madison Astronomy Club Symposium**, Madison, WI [Talk]
Aug 2023 **Lamat REU Final Presentations 2023**, Santa Cruz, CA [Talk]
Apr 2023 **UW-Madison Undergraduate Symposium**, Madison, WI [Talk]
Apr 2023 **Space Weather Workshop**, Boulder, CO [Poster]
Dec 2022 **UW-Madison Astronomy Club Symposium**, Madison, WI [Talk]
July 2022 **Boulder Solar Alliance REU Final Presentations 2022**, Boulder, CO [Talk and Poster]
Apr 2022 **UW-Madison Undergraduate Symposium**, Madison, WI [Talk]

Awards

2025 **Precandidate Rackham Graduate Student Research Grant**, University of Michigan
2024 **Chambliss Undergraduate Poster Honorable Mention**, American Astronomical Society
2024 **Graduate Research Fellowship Program**, National Science Foundation
2024 **Lowell Doherty Award for Excellence in Astronomy**, UW-Madison Department of Astronomy
2024 **Undergraduate Scholarship**, NASA Wisconsin Space Grant Consortium
2023 **Bromley Conference Travel Award**, UW-Madison L&S Honors Program
2023 **Dr. Maritza Irene Stapanian Crabtree Undergraduate Award**, UW-Madison
2023 **Best General Radio News Story**, Wisconsin Broadcasters Assoc.
2023 **Lamat Fellowship**, University of California Santa Cruz
2021-2023 **Dean's List x3**, UW-Madison
2022 **William F. Vilas Scholarship**, UW-Madison
2021-2022 **Undergraduate Research Scholars**, UW-Madison
2019-2021 **Dean's List x4**, Saint Cloud State University

Computer Skills

Languages: Python, IDL, Matlab

Projects:

GOES-R XRS (New) Flare Location Parameter Algorithm *May 2022-June 2023*
Lead Developer and Maintainer
Improves the XRS flare location calculation algorithm with a Monte Carlo routine that optimizes the geometric factors in the algorithm via comparisons with locations determined using Solar Dynamics Observatory Atmospheric Imaging Assembly. Due to export control this project is not publicly available. Data and ReadMe available at `https://www.ngdc.noaa.gov/stp/satellite/goes-r.html`

Brown Dwarf Visualizer *August 2025*
Co-Lead Developer
Created and released small package to plot brown dwarfs in an interactive plotting space during 2025 Code Astro program. See `https://github.com/astrojorge/BDViz`

Teaching

UW-Madison Physics Learning Center, Peer Mentor Tutor *September 2023-May 2024*
Group-based active learning for Physics 103, 104, 207, and 208

Observations

JWST Cycle 4 Program 8155, *Storms are Brewing in the ONC: Variability Monitoring of 50 Substellar Worlds* (43.3 hrs, PI Limbach; Co-I Kotten)

UW-Madison Pine Bluff 2.3 Meter Small Radio Telescope (ASTRON 465, 3 labs)

Peter Livingston Optical 16 inch Telescope (ASTRON 465, 2 labs)

Other Presentations

Mar 2026 **UMich Museum of Natural History, Ann Arbor, MI**
Women in Science & Engineering Residency Program invited speaker

Dec 2025 **Astronomy on Tap, Ann Arbor, MI**
"Dear Pluto" (science play) performance

Oct 2025 **University of Michigan, Ann Arbor, MI**
NSF Graduate Research Fellowships Program Q&A –Invited Panelist

May 2024 **Astronomy on Tap, Madison, WI**
Dancing Lights: Exploring Auroras from Earth to Exoplanets

Nov 2023 **Astronomy Club, Madison, WI**
Research Experience for Undergraduates and Internships Panel – Panelist

July 2023 **Badger Talks Live, Madison, WI**
Devoured Worlds: Lessons From Planet-Ingesting Stars –Invited introduction

Mar 2023 **WSUM News *On Wisconsin* Podcast, Madison, WI**
On Wisconsin podcast E20: Women in STEM interview –Invited speaker

Dec 2022 **Astronomy Club, Madison, WI**
Research Experience for Undergraduates Panel – Panelist

Activities, Service, & Outreach

Astronomy on Tap Ann Arbor, Lead Organizer *July 2025-Present*

University of Michigan Museum of Natural History, Planetarium Operator *September 2024-Present*

UW-Madison Astronomy Department, Undergraduate Representative *Februrary 2023-May 2024*

Faculty Honors Committee, Student Representative *September 2022-May 2024*

WSUM Student Radio News Team, Science Desk *February 2022-February 2024*

UW-Madison Astronomy Club, Member *September 2021-May 2024*